

# AI Bluetooth Earbuds Assistant

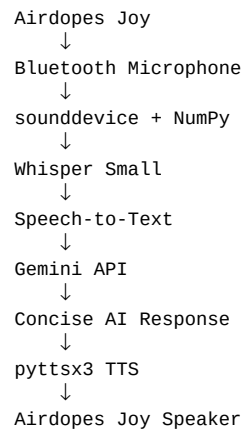
FINAL PROJECT DOCUMENTATION — Complete Voice Assistant Prototype

| Item             | Final Details                               |
|------------------|---------------------------------------------|
| Bluetooth device | Airdopes Joy                                |
| Main Python      | Python 3.14.7 — venv                        |
| STT/AI Python    | Python 3.11.9 — stt_venv                    |
| Audio stack      | sounddevice + NumPy + soundfile             |
| Testing          | pytest — Unit, Integration, Functional      |
| Speech-to-Text   | OpenAI Whisper — Small model                |
| Cloud AI         | Gemini API — gemini-3.1-flash-lite          |
| Text-to-Speech   | pyttsx3                                     |
| Local AI         | Ollama + Gemma 4 — benchmarked and removed  |
| FINAL STATUS     | <b>COMPLETE VOICE ASSISTANT LOOP — PASS</b> |

# 1. Executive Summary

Project 01 developed and validated a Bluetooth-earbud AI voice assistant prototype. The final working path is Airdopes Joy microphone → Whisper Small Speech-to-Text → Gemini AI → pyttsx3 Text-to-Speech → Airdopes Joy speaker. Bluetooth audio, automated tests, speech recognition, AI processing and voice playback were individually validated and then combined into a successful end-to-end voice loop.

# 2. Final Architecture



# 3. Development Environment

| Component         | Configuration                  |
|-------------------|--------------------------------|
| OS                | Windows / PowerShell / VS Code |
| Audio/Test Python | Python 3.14.7, venv            |
| STT/AI Python     | Python 3.11.9, stt_venv        |
| Media tool        | FFmpeg 9.0 via Winget          |

```
Activate:
.\stt_venv\Scripts\Activate.ps1

Verify:
python --version
```

# 4. Required Python Packages

| Package        | Purpose                                            |
|----------------|----------------------------------------------------|
| sounddevice    | Bluetooth microphone recording and audio playback. |
| NumPy          | Numerical audio arrays required by sounddevice.    |
| soundfile      | Read/write WAV audio files.                        |
| pytest         | Unit, integration and functional testing.          |
| openai-whisper | Local Speech-to-Text.                              |
| google-genai   | Gemini API integration.                            |
| pyttsx3        | Text-to-Speech using Windows speech engine.        |
| ollama         | Local Gemma AI evaluation.                         |

# 5. Automated Testing — Final Result

| Test                | Result                 |
|---------------------|------------------------|
| Unit Testing        | 2/2 PASS               |
| Integration Testing | 1/1 PASS               |
| Functional Testing  | 1/1 PASS               |
| ALL TESTS           | <b>4/4 PASS — 100%</b> |

```
pytest -v
```

```
4 passed, 272 warnings in 9.46s
```

All four automated tests passed. The warnings were deprecation warnings related to sounddevice/NumPy and did not cause test failures.

## 6. Bluetooth and Audio Validation

Airdopes Joy was successfully detected with a Bluetooth microphone input and stereo output. Recording and playback were successfully demonstrated.

```
[2] Airdopes Joy Hands-Free → microphone
[4] Airdopes Joy Stereo      → speaker output
```

## 7. Whisper Speech-to-Text

Whisper was installed in the Python 3.11.9 environment. Recorded WAV audio and live Bluetooth microphone audio were successfully transcribed. The project moved from the base model to Whisper Small to improve recognition.

```
result = whisper_model.transcribe(
    AUDIO_FILE,
    language="en",
    fp16=False,
    temperature=0,
    condition_on_previous_text=False
)
```

The improved configuration correctly recognized the final spoken request: **“Tell me about Bangalore.”**

## 8. Local AI Evaluation

Ollama + Gemma 4 was tested as a local AI option. The measured baseline response time was **84.65 seconds**. Gemma 4 and Gemma 4 E2B used substantial disk space and made the system slower, so both models were removed after benchmarking. Ollama was retained.

```
Gemma 4 local AI baseline: 84.65 seconds
```

## 9. Gemini AI Evaluation

Gemini was selected as the practical cloud AI backend. The API key was configured through an environment variable and was not stored in source code.

```
Standalone Gemini response time: 1.16 seconds
```

A concise-response test later measured **1.43 seconds**.

## 10. Whisper → Gemini Integration

```
Airdopes Joy
↓
Whisper Small
↓
"Tell me about India."
↓
Gemini
↓
AI response
```

| Stage       | Observed result |
|-------------|-----------------|
| Recording   | PASS — 6.19 sec |
| Whisper     | PASS — 4.79 sec |
| Gemini      | PASS — 3.85 sec |
| Integration | <b>PASS</b>     |

## 11. Text-to-Speech

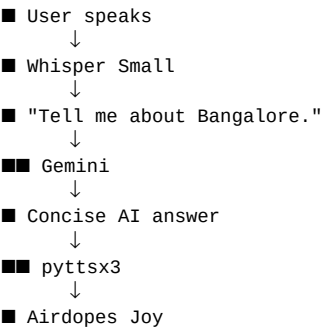
pyttsx3 was installed and tested successfully. The sample response was converted to speech and the TTS stage was validated.

Text → pyttsx3 → Windows audio output → Airdopes Joy

Observed TTS processing time: 6.97 seconds  
Result: PASS

## 12. FINAL END-TO-END VOICE ASSISTANT TEST

The final test successfully completed the entire voice loop: Airdopes Joy microphone → Whisper Small → Gemini → Text-to-Speech → Airdopes Joy speaker.



| Stage               | Final observed time | Status      |
|---------------------|---------------------|-------------|
| Bluetooth recording | 5.79 sec            | PASS        |
| Whisper STT         | 6.10 sec            | PASS        |
| Gemini AI           | 1.26 sec            | PASS        |
| Text-to-Speech      | 19.39 sec           | PASS        |
| TOTAL PIPELINE      | <b>32.59 sec</b>    | <b>PASS</b> |

## 13. Final Output

Whisper correctly transcribed **“Tell me about Bangalore.”**. Gemini generated a concise spoken-style answer about Bangalore, and the TTS stage successfully converted that answer to audio. The complete Airdopes-to-AI-to-Airdopes loop therefore passed.

## 14. Performance Analysis

| Component | Time      | Observation                   |
|-----------|-----------|-------------------------------|
| Recording | 5.79 sec  | Bluetooth capture working     |
| Whisper   | 6.10 sec  | Speech recognition processing |
| Gemini    | 1.26 sec  | Fastest AI stage              |
| TTS       | 19.39 sec | <b>Current bottleneck</b>     |

| Component | Time             | Observation                |
|-----------|------------------|----------------------------|
| Total     | <b>32.59 sec</b> | Complete voice interaction |

The 32.59-second value is a single observed run. TTS is the largest measured latency contributor and is the main future optimization target.

## 15. Security

The Gemini API key is stored in the **GEMINI\_API\_KEY** environment variable and must never be committed to source control or included in public documentation.

```
if ($env:GEMINI_API_KEY) {  
    "Gemini API key configured"  
} else {  
    "Gemini API key missing"  
}
```

## 16. Troubleshooting Lessons

| Issue                    | Resolution                                                         |
|--------------------------|--------------------------------------------------------------------|
| Python command alias     | Used the py launcher to locate installed Python versions.          |
| PowerShell activation    | Configured the required virtual-environment activation workflow.   |
| Missing NumPy            | Installed NumPy because sounddevice recording depends on it.       |
| Separate STT environment | Used Python 3.11.9 to isolate Whisper/AI dependencies.             |
| FFmpeg PATH              | Restarted PowerShell after Winget modified PATH.                   |
| Whisper recognition      | Moved from base to Small and added English/deterministic settings. |
| Gemma resource usage     | Benchmarked and removed the large Gemma models.                    |

## 17. Final Project Status

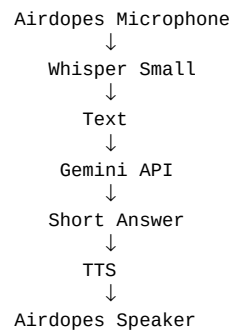
| Milestone                    | Status                      |
|------------------------------|-----------------------------|
| Bluetooth detection          | ■ COMPLETED                 |
| Microphone recording         | ■ COMPLETED                 |
| Stereo playback              | ■ COMPLETED                 |
| Automated testing            | ■ 4/4 PASS                  |
| Whisper STT                  | ■ COMPLETED                 |
| Whisper accuracy improvement | ■ COMPLETED                 |
| Local AI benchmark           | ■ COMPLETED; models removed |
| Gemini API                   | ■ COMPLETED                 |
| Concise AI response          | ■ COMPLETED                 |
| Text-to-Speech               | ■ COMPLETED                 |
| Complete voice loop          | ■ COMPLETED — PASS          |

## 18. Future Enhancements

- Optimize TTS latency.
- Evaluate a faster or streaming TTS engine.

- Add voice activity detection.
- Add wake-word or push-to-talk activation.
- Add conversation history.
- Handle Bluetooth disconnect/reconnect.
- Handle Gemini network/API failures.
- Run repeated latency benchmarks.
- Add end-to-end regression tests.
- Evaluate battery and long-duration reliability.

## 19. Final Architecture for Future Work



## 20. Useful Commands

```
# Activate
.\stt_venv\Scripts\Activate.ps1

# Automated tests
pytest -v

# Complete voice assistant
python .\voice_assistant_test.py

# Whisper → Gemini
python .\whisper_gemini_test.py

# TTS
python .\tts_test.py

# FFmpeg
ffmpeg -version
```

## 21. Final Conclusion

### PROJECT 01 — COMPLETE VOICE ASSISTANT PROTOTYPE: PASS.

The project successfully progressed from Bluetooth audio validation to a complete working AI voice assistant. Airdopes Joy provides the microphone and speaker interface, Whisper Small performs local Speech-to-Text, Gemini provides the AI response, and pytsx3 provides Text-to-Speech. The final end-to-end test passed with a measured total pipeline time of **32.59 seconds**.

The project is ready for documentation closure and transition to the next project. TTS latency is documented as the primary future optimization area.

## 22. Final Revision Record

| Revision   | Update                                                                                                                 |
|------------|------------------------------------------------------------------------------------------------------------------------|
| FINAL      | Added complete voice-assistant implementation and final end-to-end test result.                                        |
| Final test | Bluetooth Microphone PASS; Audio Recording PASS; Whisper STT PASS; Gemini AI PASS; TTS PASS; Airdopes Voice Loop PASS. |

| Revision        | Update                                                                                 |
|-----------------|----------------------------------------------------------------------------------------|
| Final timing    | 5.79 sec recording; 6.10 sec Whisper; 1.26 sec Gemini; 19.39 sec TTS; 32.59 sec total. |
| Final milestone | <b>Airdopes → Whisper → Gemini → TTS → Airdopes successfully demonstrated.</b>         |